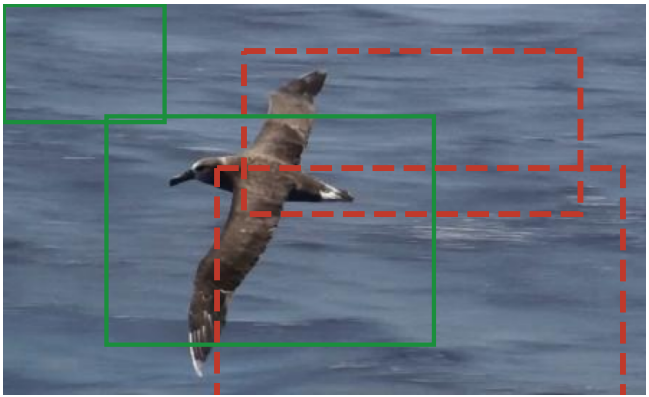
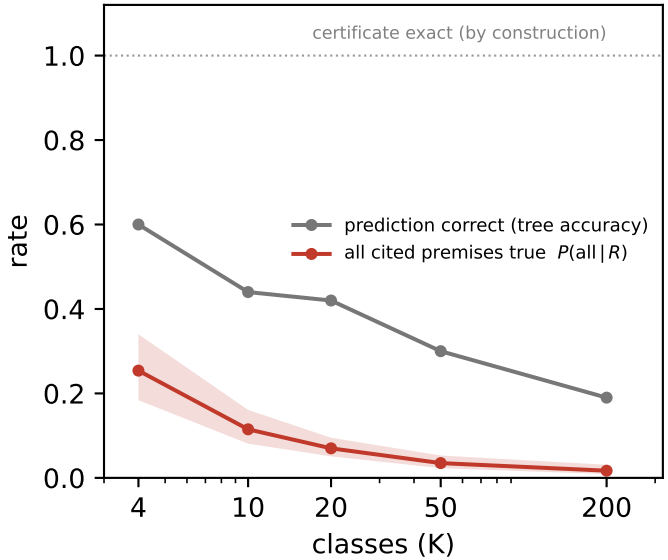

where the proof stops
image
CUB, oracle part boxes
DINOv3 patches
frozen
predicates $\phi(x)$
$F = 36$ part-bound literals
logistic readers
decision tree
CART $d = 12$, predicted bits
Setting T / distilled D
sufficient reason R
$R \models c$, minimal
engine cross-check
✗
✗ ✓ ✓
the five reason bits (sid 44)
✗
measured — learned from the image
exact — by construction




certificate exact (by construction)
1.0
0.8
0.6
0.4
0.2
0.0
rate
prediction correct (tree accuracy)
all cited premises true $P(\text{all} \mid R)$
4
10
20
50
200
classes (K)